

Housing Price Prediction Project Summary

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This project focused on analyzing a housing dataset and building machine learning models to predict house prices. The dataset was loaded and explored using Pandas, and data preprocessing techniques were applied to prepare the data for modeling. Categorical features were converted into numerical form, duplicate records were checked, and the dataset was organized into input features and a target variable (price).

Several visualizations were created to better understand the data. A histogram showed the distribution of house prices, a correlation heatmap highlighted relationships between features and price, and a scatter plot demonstrated how house area influences property value. These visualizations helped identify the most important factors affecting house prices.

Two machine learning models were trained and evaluated: Linear Regression and Random Forest Regressor. Their performance was measured using MAE, RMSE, and R^2 Score. The models were able to predict house prices with reasonable accuracy, showing that property characteristics such as area, number of bathrooms, bedrooms, parking availability, and other amenities play an important role.

One interesting observation was that houses with a similar number of bedrooms could have very different prices, indicating that size and additional facilities strongly influence value. Based on the findings, real estate businesses should focus on properties with larger living areas and modern amenities, as these features are more likely to increase market value and attract buyers.